

TECHNICAL PROJECT REPORT: CLINICAL DECISION SUPPORT SYSTEM

Integration of YOLOv8 Object Detection, FastAPI Backend, Interactive Frontend, and Embedded Hardware Systems

1. EXECUTIVE SUMMARY & PROJECT OBJECTIVE

This project introduces an interactive **Clinical Decision Support System (CDSS)** designed to assist laboratory personnel and physicians in identifying, counting, and analyzing blood components (**RBCs, WBCs, Platelets**) and bacterial strains directly from microscopic slides.

The primary objective is not merely displaying marked images, but generating quantitative data cross-referenced with a medical knowledge base to validate treatment plans proposed by attending doctors.

2. YOLOV8 MODEL OUTPUT ARCHITECTURE

The custom-trained YOLOv8 model (best.pt) ingests raw microscope imagery and delivers two synchronized layers of diagnostic output:

- **Visual Annotation Output:** Overlays color-coded bounding boxes around detected micro-organisms, complete with class identification tags and probability confidence scores (e.g., *Salmonella* 0.94).
- **Quantitative Aggregate Output:** Extracts raw coordinates into exact numerical totals (e.g., RBC: 12, WBC: 4, E-coli: 3), eliminating manual counting errors.

3. FASTAPI BACKEND & WORKFLOW PIPELINE

The FastAPI backend acts as the decision core connecting model inference, database rule verification, and client interfaces via a two-stage interactive workflow:

Phase A: Automated Sample Analysis (POST /api/v1/analyze)

1. Ingests raw microscopic image files streamed from the front end or hardware camera.
2. Runs YOLOv8 inference to aggregate entity counts and bounding box geometries.
3. Queries the **Clinical Rule Database** to compare identified counts against standard physiological reference ranges (*threshold limits*).
4. Returns a JSON payload containing statistical breakdowns and preliminary diagnostic warnings.

Phase B: Physician Treatment Validation (POST /api/v1/validate-medication)

1. Accepts the specific drug prescription entered by the attending physician (Doctor's Proposed Medication).

2. Cross-references the proposed drug against approved pharmaceutical guidelines mapped to the detected pathogen in the database.
3. Issues a definitive validation status (**APPROVED** vs. **WARNING / MISMATCH**) alongside recommendations for secondary confirmatory tests (e.g., *Widal Test*, *Stool Culture*).

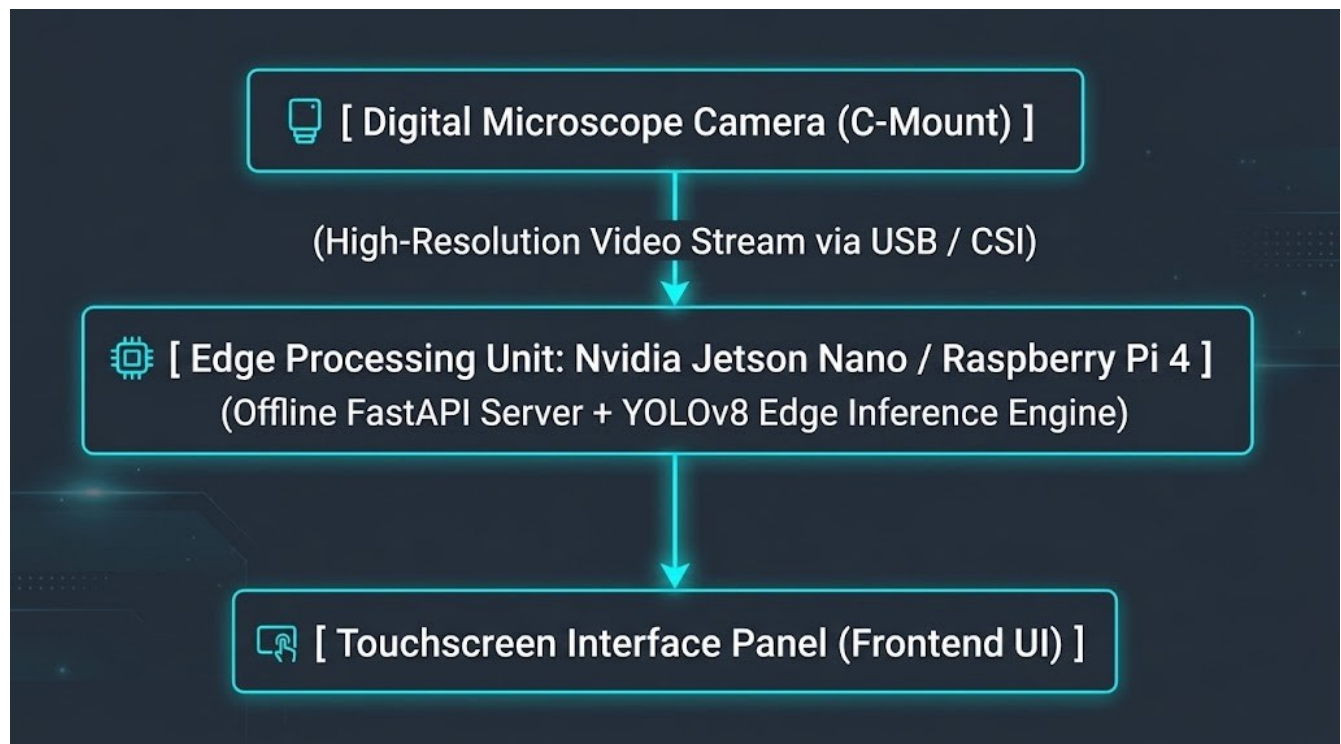
4. FRONTEND USER INTERFACE DESIGN

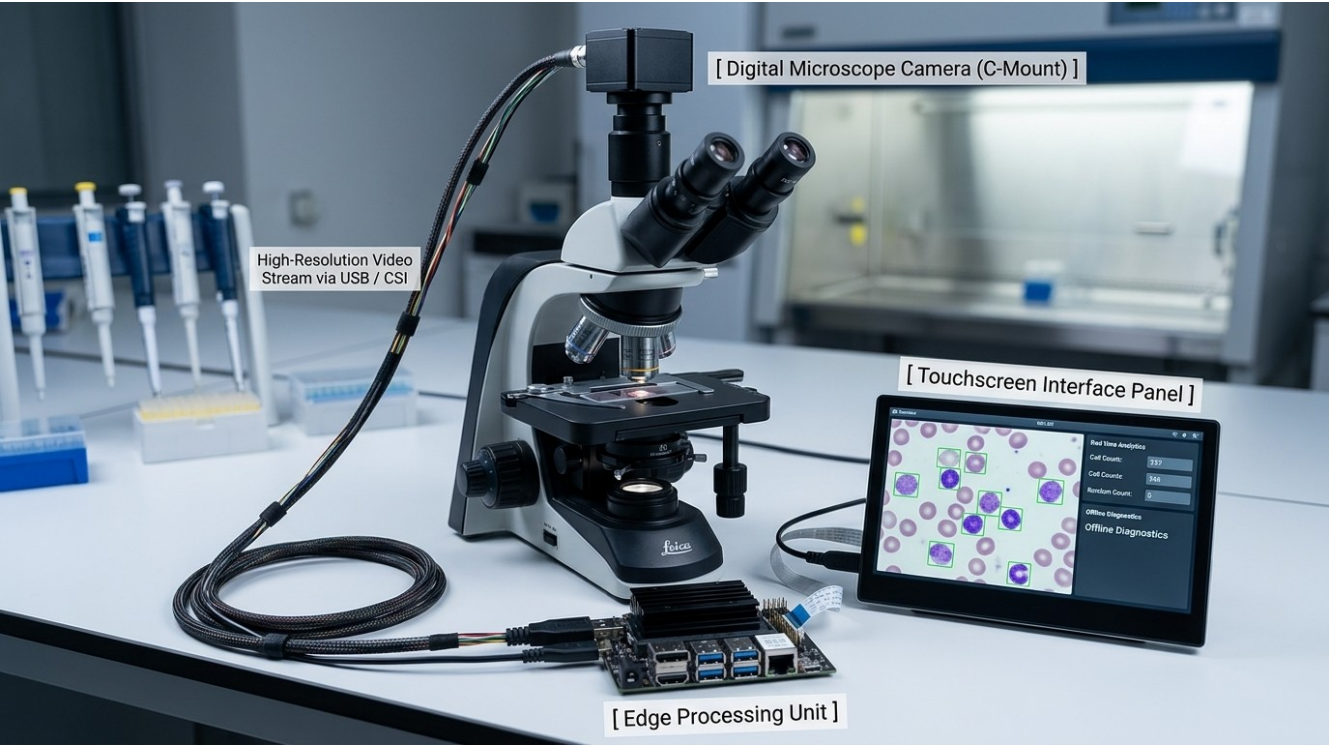
The user interface provides a dual-column layout tailored for clinical decision-making:

- **Live Feed & Image Ingestion Panel:** Displays live optical video streams directly from the microscope camera or allows manual slide image uploads.
- **Annotated Overlay Display:** Renders the source image with responsive bounding boxes provided by the vision engine.
- **Quantitative Statistics Dashboard:** Displays real-time numerical tallies and distribution metrics for each detected cell and pathogen class.
- **Interactive Clinical Action Panel:** Contains an input field for the doctor's intended medication plan and a "**Validate Treatment**" button that displays immediate green/red feedback from the system backend.

5. EMBEDDED HARDWARE INTEGRATION & ARCHITECTURE

To operate as a standalone diagnostic Point-of-Care (PoC) laboratory unit, the system integrates vision models and backend execution onto specialized edge hardware:





Hardware Components & Responsibilities:

- 1. **Digital Microscope C-Mount Camera:** Mounted directly onto standard laboratory optical microscopes to capture high-definition slide samples.
- 2. **Embedded Edge Processor (Nvidia Jetson Nano / Raspberry Pi 4):**
 - **Nvidia Jetson Nano:** Preferred deployment platform due to integrated CUDA GPU cores, enabling real-time local model inference without requiring cloud network connectivity.
 - **Raspberry Pi 4:** Supported fallback platform utilizing quantized lightweight models (ONNX or TFLite runtimes).
- 3. **Integrated Touchscreen Display:** Provides an intuitive touch-driven interface for clinicians to view diagnostic dashboards and input prescription parameters directly at the benchtop.

Hardware Components and Cost Breakdown

No.	Hardware Component	Function & Specifications	Estimated Cost (USD)	Estimated Cost (TZS)
1	Digital Microscope Camera (C-	Mounts onto the microscope to capture high-resolution image/video	\$80 – \$250	TZS 210,000 – 650,000

No.	Hardware Component	Function & Specifications	Estimated Cost (USD)	Estimated Cost (TZS)
	Mount)	streams for processing.		
2	Edge Processing Unit (<i>Nvidia Jetson Nano / Raspberry Pi 4</i>)	Micro-computer hosting the offline FastAPI server and AI model (YOLOv8 Edge Inference Engine).	\$75 – \$150	TZS 195,000 – 390,000
3	Touchscreen Interface Panel	Capacitive touchscreen (7" or 10.1") displaying the Frontend UI, live stream, and AI diagnostic results.	\$40 – \$90	TZS 105,000 – 235,000
4	System Accessories & Cables	High-speed USB/CSI data cables, HDMI, power adapter (5V/4A), and protective casing.	\$20 – \$40	TZS 52,000 – 105,000
TOTAL	Overall Project Hardware Cost	Minimum to Maximum Estimated Budget	\$215 – \$530	TZS 560,000 – 1,380,000

6. CONCLUSION & FUTURE DEVELOPMENTS

By unifying computer vision (YOLOv8), high-throughput web APIs (FastAPI), interactive clinician workflows, and edge computing hardware, this system delivers an offline, reliable diagnostic support solution. Future milestones will focus on conducting edge hardware speed benchmarking and integrating with Hospital Information Management Systems (HIMS).